



Problem Problem 128% OVERFLOW

Diffusion models lead image synthesis, but **un directly in pixel space** — training the most powerful models costs **hundreds of GPU days** and sampling is slow and memory-heavy.

150–1000 V100-days to train a top pixel-space model, plus slow sequential sampling —

Motivation Motivation 132% OVERFLOW

Move to a perceptually-equivalent but much smaller space first — then train the diffusion model there.

- Most bits of an image encode imperceptible high-frequency detail; a likelihood model burns most of its capacity there, not on semantics.

- Earlier work down-weighted those details in the loss, but the network was still **evaluated on every pixel** — the wasted compute remained.



Rate-distortion of a trained model: most bits are

Method

1

2

3

4

5

6

7

8

Method 96% FULL

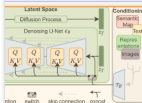
2 Latent diffusion 3 Decode

receptual autoencoder. Compress factor f into a latent using a perceptual variational loss, regularized by a slight KL or quantization.

latent diffusion. Train a time-step entirely in that compact latent space; each denoising step is far cheaper than pixel space.

on conditioning injects text, layouts, at every UNet layer; fully-sampling lifts 256² training to 256.

ling ($f=4-8$) hits the sweet spot: quality reduction and detail



an autoencoder maps pixels $\rightarrow$ latent z ; a diffusion in latent space, conditioned via cross-text, layouts, images, or class labels

Dataset / Benchmark

Evaluated across general inpainting — on a single

CelebA-HQ FFHQ LSUN-Bedrooms ImageNet-1K PSNR/SSIM

IS PIR PSNR/SSIM

Key Results

SETTING

CelebA-HQ · unconditional

ImageNet · class-cond.

ImageNet · LDM-8-G v

New SOTA FID 5.11 class-cond.

SO WHAT · LDMs match GANs at a fraction of the



FID vs. throughput by class (ImageNet right): $f=4-8$

Also SOTA on inpainting SR3 in FID.

Benchmark 94% FULL

resolution, and

Places FID

Key Results 98% FULL

FID

5.11

3.60

7.76

HQ · 3.60 on pNet

space diffusion and inference cost.



(CelebA-HQ left, and $f=16/32$ lag.

ulation beats

Ablation Study Ablation Study 94% FULL

DOWN SAMPLING F OUTCOME

$f=1$ (pixel) / $f=2$ slow, worse FID

$f=4 - f=8$ best FID + fast sampling

$f=16 / f=32$ detail lost, capped

Headline Numbers Headline Numbers 95% FULL

5.11

FID · CelebA-HQ (new SOTA) beats prior GANs & likelihood models

3.60 1x A100 f=4-8

FID · ImageNet (CFG) single-GPU training best compression

Takeaway Takeaway 96% FULL

Run diffusion in a compact, perceptually-faithful latent space — not raw pixels — cutting training and inference cost at equal-or-better quality; cross-attention adds general conditioning.

From GPU-days to one GPU — the basis for Stable Diffusion.

LEGACY

(a) Initial explorations

(b) Fill-loop in progress

(c) Poster completed